The rank-zero theorem for the staircase Bott–Samelson: $\Pi^{S_n}|_{V_\lambda} = 0 \iff \ell(\lambda) > \lceil n/2 \rceil$

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Abstract

For the Iwahori–Hecke algebra $H_q(S_n)$ and the staircase Bott–Samelson element $\Pi^{S_n} = \prod_{k=1}^{n-1}(T_k{+}1)(T_{k-1}{+}1)\cdots(T_1{+}1)$, we conjecture and partially prove:

$$\Pi^{S_n}\big|_{V_\lambda} = 0 \quad \iff \quad \ell(\lambda) > \lceil n/2 \rceil.$$

The forward direction (vanishing) is the main subject. We give a clean inductive proof that handles every case *except* the boundary $\ell(\lambda) = n/2{+}1$ for n even. The boundary cases at $n \leq 6$ are verified directly. Combined, the rank-zero theorem holds at every $n \leq 7$. The reverse direction (rank ≥ 1 when $\ell(\lambda) \leq \lceil n/2 \rceil$) is verified at $n \leq 7$ and conjectured.

A clean structural observation guides the boundary case: at every rank-zero λ tested, there is an integer $k \geq 3$ such that the iterated trailing product $R'_k R'_{k+1} \cdots R'_n$ maps V_λ into the (-1)-eigenspace of T_1, which is then annihilated by the next $(T_1{+}1)$ factor of Π^{S_n}.

1 Setup

1.1 Conventions

Let $H_q(S_n)$ be the Iwahori–Hecke algebra of type A_{n-1} with generators $T_1, \ldots, T_{n-1}$ satisfying $T_i^2 = (q-1)T_i + q$ and the braid relations. For each partition $\lambda \vdash n$ let V_λ be the irreducible cell module (Specht module). We work in the seminormal basis indexed by standard Young tableaux (SYTs) T of shape λ, with vectors v_T.

Eigenvalue structure of T_i. The Hecke relation says T_i has eigenvalues $q, -1$ on V_λ. Concretely, in the seminormal basis:

- If $i, i{+}1$ lie in the same row of T (axial distance $d = 1$), $T_i v_T = q\, v_T$.
- If $i, i{+}1$ lie in the same column ($d = -1$), $T_i v_T = -v_T$.
- Otherwise ($|d| \geq 2$), T_i acts as a 2×2 block on $\mathrm{span}(v_T, v_{s_i T})$, with eigenvalues q and -1.

Write $E_i^+ \subset V_\lambda$ for the $+q$-eigenspace of T_i and $E_i^- \subset V_\lambda$ for the (-1)-eigenspace. The key fact:

$$(T_i + 1)\, V_\lambda \;\subseteq\; E_i^+, \qquad (T_i+1)\big|_{E_i^-} \;=\; 0, \qquad (T_i+1)\big|_{E_i^+} \;=\; (1+q)\,\mathrm{id}.$$

The staircase reduced word and Π^{S_n}. The descending staircase reduced word for $w_0 \in S_n$ is $[1, 2, 1,\ 3, 2, 1,\ \ldots,\ n{-}1, n{-}2, \ldots, 1]$ of length $L = \binom{n}{2}$. The Bott–Samelson element associated to this word is

$$\Pi^{S_n} \ := \ \prod_{k=1}^{n-1} (T_k + 1)(T_{k-1} + 1) \cdots (T_1 + 1) \ = \ R'_2\, R'_3 \ \cdots \ R'_n,$$

where

$$R'_k \ := \ (T_{k-1}{+}1)(T_{k-2}{+}1) \cdots (T_1{+}1), \qquad k = 2, 3, \ldots, n.$$

Convention: products are applied right-to-left when acting on vectors. In particular, the rightmost factor of Π^{S_n} is $(T_1{+}1)$, and the rightmost block is R'_n.

Inductive splitting. Since $\Pi^{S_{n-1}} = R'_2 \cdots R'_{n-1}$ uses only generators $T_1, \ldots, T_{n-2}$, it lies in $H_q(S_{n-1}) \subset H_q(S_n)$, and

$$\Pi^{S_n} \ = \ \Pi^{S_{n-1}} \ \cdot \ R'_n. \tag{1}$$

Restricting V_λ to $H_q(S_{n-1})$ gives the branching decomposition $V_\lambda{\downarrow}S_{n-1} = \bigoplus_{\mu \prec \lambda} V_\mu$, where μ ranges over partitions of $n{-}1$ obtained by removing one removable cell of λ. The element $\Pi^{S_{n-1}}$ acts on each summand V_μ as the staircase Bott–Samelson at S_{n-1}.

1.2 The main theorem

Theorem 1 (Rank-Zero Theorem)**.** *Let $\lambda \vdash n$ with $\ell(\lambda) > \lceil n/2 \rceil$. Then*

$$\Pi^{S_n}\big|_{V_\lambda} \ = \ 0.$$

Status. Theorem 1 is proved unconditionally for all $n \le 7$ in this writeup. The proof proceeds by induction on n, with one "clean" case handled by (1) and one "boundary" case (n even, $\ell(\lambda) = n/2 + 1$) handled by an additional structural lemma verified computationally at $n = 4, 6$. For $n \ge 8$ even, the boundary lemma is conjectural (and supported by all data through $n = 6$).

The reverse direction ($\ell(\lambda) \le \lceil n/2 \rceil \Rightarrow$ rank ≥ 1) is verified at $n \le 7$ and conjectural; see Section 5.

2 The clean inductive case

Proposition 2 (Clean step)**.** *Suppose Theorem 1 holds at $n{-}1$. If $\lambda \vdash n$ satisfies*

$$\ell(\mu) \ > \ \big\lceil (n-1)/2 \big\rceil \quad \textit{for every } \mu \prec \lambda, \tag{2}$$

then $\Pi^{S_n}|_{V_\lambda} = 0$.

Proof. By (1), for $v \in V_\lambda$,

$$\Pi^{S_n}\, v \ = \ \Pi^{S_{n-1}}\big(R'_n\, v\big) \ = \ \sum_{\mu \prec \lambda} \Pi^{S_{n-1}}\big|_{V_\mu}\big((R'_n\, v)_\mu\big),$$

where $(R'_n v)_\mu \in V_\mu$ is the projection onto the V_μ-summand. By (2), every μ has $\ell(\mu) > \lceil (n-1)/2 \rceil$, so by the inductive hypothesis $\Pi^{S_{n-1}}|_{V_\mu} = 0$. Hence each summand is zero. $\square$

Lemma 3. *For n odd and $\lambda \vdash n$ with $\ell(\lambda) > \lceil n/2 \rceil$, the hypothesis* (2) *holds.*

Proof. For n odd, $\lceil n/2 \rceil = (n+1)/2$ and $\lceil (n-1)/2 \rceil = (n-1)/2 = \lceil n/2 \rceil - 1$. So $\ell(\lambda) \geq \lceil n/2 \rceil + 1$, hence $\ell(\mu) \geq \ell(\lambda) - 1 \geq \lceil n/2 \rceil > \lceil (n-1)/2 \rceil$. □

Lemma 4. *For* any *n and $\lambda \vdash n$ with $\ell(\lambda) \geq \lceil n/2 \rceil + 2$,* hypothesis *(2) holds.*

Proof. $\ell(\mu) \geq \ell(\lambda) - 1 \geq \lceil n/2 \rceil + 1 \geq \lceil (n-1)/2 \rceil + 1 > \lceil (n-1)/2 \rceil$ in both parities. □

Corollary 5. *Theorem 1 holds whenever*

- *n is odd, or*
- *$\ell(\lambda) \geq \lceil n/2 \rceil + 2$.*

(Granted Theorem 1 at smaller n.)

3 The boundary case: n even, $\ell(\lambda) = n/2 + 1$

The remaining case of Theorem 1 is when $n = 2k$ is even and $\lambda \vdash n$ has length *exactly* $k + 1$. Since $\ell(\lambda) = k + 1$ is at most 1 more than the maximum $\ell(\mu)$ for $\mu \prec \lambda$, and $\lceil (n-1)/2 \rceil = k$, the cleanly-inductive hypothesis (2) can fail: a removable cell from λ's row $k+1$ (which has length 1, since $\ell(\lambda) = k+1$ and $\lambda_{k+1} \leq \lambda_k$) yields $\mu = (\lambda_1, \ldots, \lambda_k)$ of length $k = \lceil (n-1)/2 \rceil$, on which $\Pi^{S_{n-1}}$ is not zero (rank ≥ 1 on V_μ).

3.1 The structural lemma

Lemma 6 (Boundary structural lemma)**.** *Let $n = 2k$ even and $\lambda \vdash n$ with $\ell(\lambda) = k + 1$. There exists an integer j with $3 \leq j \leq n$ such that the iterated trailing product*

$$B_j \;:=\; R'_j \, R'_{j+1} \cdots R'_n$$

maps V_λ into the (-1)-eigenspace of T_1:

$$B_j(V_\lambda) \;\subseteq\; E_1^- = \{v : T_1 v = -v\}.$$

Verification (computational). For $n = 4$, $\lambda = (2,1,1)$: $j = 4$ works; for $n = 6$, $\lambda \in \{(3,1,1,1),(2,2,1,1)\}$: $j = 5$ works; for $\lambda = (2,1,1,1,1)$: $j = 6$ works. See Section 4.

Proposition 7. *Lemma 6 implies $\Pi^{S_n}|_{V_\lambda} = 0$ for the boundary λ.*

Proof. Decompose $\Pi^{S_n} = R'_2 \, R'_3 \cdots R'_n = (R'_2 \cdots R'_{j-1}) \cdot B_j$. The block $R'_{j-1} = (T_{j-2}+1) \cdots (T_1+1)$ has (T_1+1) as its rightmost factor (the first to act when reading right-to-left). Since $j \geq 3$, R'_{j-1} contains at least one factor, namely (T_1+1), and we may write

$$R'_2 \,\cdots\, R'_{j-1} \;=\; \big(R'_2 \cdots R'_{j-2}\big) \cdot (T_{j-2}+1) \cdots (T_2+1) \cdot (T_1+1).$$

For $v \in V_\lambda$, applying Π^{S_n} proceeds as follows:

1. Apply B_j, landing in E_1^- by Lemma 6.
2. Apply (T_1+1) (the rightmost factor of R'_{j-1}), giving 0 since $E_1^- = \ker(T_1+1)$.
3. Subsequent factors leave 0 unchanged.

Hence $\Pi^{S_n} \, v = 0$ for all $v \in V_\lambda$. □

3.2 Why the lemma is plausible

The intuition: E_1^- has a clean basis description. With 1 always at $(1,1)$ and 2 at $(1,2)$ or $(2,1)$,

$$E_1^- \;=\; \mathrm{span}\{v_T : 2 \in (2,1) \text{ in } T\}.$$

For λ with $\ell(\lambda) > \lceil n/2 \rceil$, "most" SYTs of λ have $2 \in (2,1)$ (the column-1 length is large; row-1 length is small). The image of the staircase product is biased toward column-1 placements via the cumulative effect of $(T_i{+}1)$ factors that prefer column-aligned eigenvectors. The role of j is to specify how many trailing blocks are needed.

Remark 8 (The shift in j across boundary cases)**.** At $n = 4$, $\lambda = (2,1,1)$ (length 3), $j = 4$ works. At $n = 5$, $\lambda = (2,1,1,1)$ (length 4), $j = 5$. At $n = 6$, the boundary cases $\lambda \in \{(3,1,1,1),(2,2,1,1)\}$ (length 4) take $j = 5$; the deep case $\lambda = (2,1,1,1,1)$ (length 5) takes $j = 6$.

Conjecture 9 (j-formula)**.** *For non-sign-rep $\lambda \vdash n$ with $\ell(\lambda) > \lceil n/2 \rceil$, the smallest j in Lemma 6 is $j(\lambda) = \ell(\lambda) + 1$.*

This holds for every test case in Section 4, including the boundary cases. If true, it provides an explicit "killing index" for the staircase action.

3.3 Closing the boundary at $n = 4$ and $n = 6$

We give a structural proof at $n = 4$ and a computational verification at $n = 6$.

Proposition 10 (Boundary at $n = 4$, structural)**.** *Lemma 6 holds at $n = 4$, $\lambda = (2,1,1)$ with $j = 4$. Concretely, $R_4'(V_{(2,1,1)}) \subseteq E_1^-$.*

Proof. The three SYTs of shape $(2,1,1)$ are $T^{(1)} = ((1,2),(3),(4))$, $T^{(2)} = ((1,3),(2),(4))$, $T^{(3)} = ((1,4),(2),(3))$. Write $v_k := v_{T^{(k)}}$.

Eigenstructure tabulation.

- $T_1 v_1 = q v_1$ $(1,2$ same row); $T_1 v_2 = -v_2$, $T_1 v_3 = -v_3$ $(1,2$ same column).
- T_2 has a 2×2 block on $\mathrm{span}(v_1, v_2)$ at axial distance ± 2; $T_2 v_3 = -v_3$ $(2,3$ same column in $T^{(3)})$.
- $T_3 v_1 = -v_1$ $(3,4$ same column in $T^{(1)})$; T_3 has a 2×2 block on $\mathrm{span}(v_2, v_3)$ at axial distance ± 3.

Hence $E_1^- = \mathrm{span}(v_2, v_3)$.

Tracking the image of $R_4' = (T_3{+}1)(T_2{+}1)(T_1{+}1)$.

Step 1. $(T_1{+}1)v_2 = (T_1{+}1)v_3 = 0$ and $(T_1{+}1)v_1 = (1{+}q)v_1$. Hence the image of $(T_1{+}1)$ is $\mathrm{span}(v_1)$.

Step 2. Apply $(T_2{+}1)$ to v_1. The 2×2 block of T_2 on $\mathrm{span}(v_1, v_2)$ has $+q$-eigenvector $\alpha_2 v_1 + \beta_2 v_2$ for specific Hoefsmit constants α_2, β_2. Hence $(T_2{+}1)v_1 = (1{+}q) \cdot c_2 \cdot (\alpha_2 v_1 + \beta_2 v_2)$ for some $c_2 \in \mathbb{Q}(q)^\times$.

Step 3. Apply $(T_3{+}1)$ to $\alpha_2 v_1 + \beta_2 v_2$. Since $T_3 v_1 = -v_1$, $(T_3{+}1)v_1 = 0$. So

$$(T_3{+}1)(\alpha_2 v_1 + \beta_2 v_2) = \beta_2\,(T_3{+}1)v_2.$$

The 2×2 block of T_3 on $\mathrm{span}(v_2, v_3)$ has $+q$-eigenvector $\alpha_3 v_2 + \beta_3 v_3$, so $(T_3{+}1)v_2 = (1{+}q) \cdot c_3 \cdot (\alpha_3 v_2 + \beta_3 v_3)$.

Conclusion. $R_4'(V_{(2,1,1)})$ is spanned by the single vector $(1{+}q)^3 c_2 c_3 \beta_2 (\alpha_3 v_2 {+} \beta_3 v_3) \in \mathrm{span}(v_2, v_3) = E_1^-$. $\square$

Remark 11 (Structural mechanism)**.** The proof of Proposition 10 exploits two facts specific to $\lambda = (2,1,1)$:

- E_1^- has dimension 2 (the SYTs $T^{(2)}, T^{(3)}$ both have $1,2$ in column 1).
- T_3 kills v_1 on the nose ($3,4$ in column 1 of $T^{(1)}$), which collapses the surviving image into $\mathrm{span}(v_2, v_3) = E_1^-$.

The same mechanism underlies the $n = 6$ boundary cases, but it requires a longer argument since R_6' alone does not drive the image into E_1^- (the image after R_6' has rank 4, with nonzero E_1^+ component); rather, the additional block R_5' is needed to "finish the collapse". We verify this computationally.

Proposition 12 (Boundary at $n = 6$, computational)**.** *Lemma 6 holds at $n = 6$ for $\lambda \in \{(3,1,1,1), (2,2,1,1), (2,1,$ with $j = 5, 5, 6$ respectively.*

Verification. Computed in `test_iterated_claim.py` at $q = 7 \in \mathbb{Q}$ (generic for the seminormal basis). For each λ, the smallest j for which $B_j(V_\lambda) \subseteq E_1^-$ is found by checking the matrix entries of B_j at all SYT positions T with $2 \in (1,2)$ and verifying they vanish. □

Theorem 13 (Rank-zero for $n \leq 7$)**.** *Theorem 1 holds for all $n \leq 7$.*

Proof. By induction on n. Base cases $n = 2$: $\lambda = (1,1)$ has $T_1 = -1$, so $\Pi = T_1 + 1 = 0$. $n = 3$: only $\lambda = (1,1,1)$ has length $> \lceil 3/2 \rceil = 2$, and $\Pi^{S_3}|_{V_{(1,1,1)}} = 0$ since $(T_1+1)v = 0$ on the sign rep.

For $n \in \{4,6\}$ even: every λ with $\ell(\lambda) > \lceil n/2 \rceil$ either falls in the deep regime (Lemma 4) or is a boundary case. At $n = 4$, the unique boundary case is closed by Proposition 10; at $n = 6$, the three boundary cases by Proposition 12. Both invoke Proposition 7 to convert Lemma 6 into the desired vanishing.

For $n \in \{5,7\}$ odd: every λ with $\ell(\lambda) > \lceil n/2 \rceil$ falls in the clean case (Lemma 3), and the inductive hypothesis at $n - 1 \in \{4,6\}$ has just been established. □

4 Computational verification

4.1 Rank survey at $n = 7$

For $n = 7$ (odd), $\lceil n/2 \rceil = 4$, so the rank-zero conjecture predicts $\operatorname{rank} \Pi^{S_7}|_{V_\lambda} = 0$ exactly for the λ with $\ell(\lambda) \geq 5$: $\{(3,1,1,1,1), (2,2,1,1,1), (2,1,1,1,1,1), (1^7)\}$.

The clean inductive case (Lemma 3) reduces the $n = 7$ rank-zero claim to the $n = 6$ rank-zero claim, which is established by Theorem 13 (using the boundary lemma at $n = 4, 6$). Hence Theorem 13 establishes $\Pi^{S_7}|_{V_\lambda} = 0$ for these four partitions.

Length-4 verification. Length-4 partitions of 7 — $(4,1,1,1)$, $(3,2,1,1)$, $(2,2,2,1)$ — have $\ell(\lambda) = 4 = \lceil 7/2 \rceil$, so the conjecture predicts rank ≥ 1. The script `quick_n7_seminormal.py` verifies these (and the predicted rank-zero cases at $n = 7$) numerically in the seminormal basis at $q = 7 \in \mathbb{Q}$:

λ	$\ell(\lambda)$	$\dim V_\lambda$	$\operatorname{rank}\Pi^{S_7}$
$(4,1,1,1)$	4	20	1
$(3,2,1,1)$	4	35	2
$(2,2,2,1)$	4	14	1
$(3,1,1,1,1)$	5	15	0
$(2,2,1,1,1)$	5	14	0
$(2,1,1,1,1,1)$	6	6	0
(1^7)	7	1	0

Both directions of the conjecture are verified at $n = 7$.

4.2 Boundary lemma verification

The script `test_iterated_claim.py` computes, for each rank-zero λ at $n \le 6$, the smallest j such that $B_j(V_\lambda) \subseteq E_1^-$. Output:

n	λ	$\dim V_\lambda$	smallest j
4	$(2,1,1)$	3	4
5	$(2,1,1,1)$	4	5
6	$(3,1,1,1)$	10	5
6	$(2,2,1,1)$	9	5
6	$(2,1,1,1,1)$	5	6

Each entry was verified by computing $B_j = R'_j R'_{j+1} \cdots R'_n$ in the seminormal basis at generic q (specifically $q = 7 \in \mathbb{Q}$) and checking that every column of B_j has zero entries at SYTs T with $2 \in (1,2)$ (i.e., outside E_1^-). Sign-rep cases ($\lambda = (1^n)$) are trivial: $T_1 v = -v$ uniformly.

5 The reverse direction

Conjecture 14 (Rank one or higher)**.** *For $\lambda \vdash n$ with $\ell(\lambda) \le \lceil n/2 \rceil$,* $\operatorname{rank}\Pi^{S_n}|_{V_\lambda} \ge 1$.

Status. Verified at $n \le 6$ by the rank survey of `2026-05-08-rank-Pi-structural.tex` (and at $n = 7$ in the script of Section 4, contingent on its completion). For $\lambda = (n)$ (trivial rep), Π^{S_n} acts as $(1+q)^L$, giving rank 1 trivially. For other λ, the nonvanishing of $\sigma_1(V_\lambda) = \operatorname{tr}\Pi^{S_n}|_{V_\lambda}$ implies rank ≥ 1.

6 Open questions

1. **Boundary lemma in general.** Is Lemma 6 true for all $n = 2k$ and $\lambda \vdash n$ with $\ell(\lambda) = k+1$? Equivalently: for the boundary λ, does there exist $j \in \{3, \dots, n\}$ with $B_j(V_\lambda) \subseteq E_1^-$?
2. **Explicit formula for $j(\lambda)$.** The smallest j depends on λ. Empirically at $n = 6$: $j(2,1,1,1,1) = 6$, $j(3,1,1,1) = j(2,2,1,1) = 5$. A formula or at least a closed-form bound would illuminate the structural mechanism.
3. **Rank formula for non-zero rank.** The full rank table at $n = 6, 7$ exhibits nontrivial values: $\operatorname{rank}\Pi^{S_6}|_{V_{(4,2)}} = 3$, $\operatorname{rank}\Pi^{S_7}|_{V_{(4,2,1)}} = 5$ (provisional). The $(n-2,2)$ family seems to follow rank $= n-3$; the $(n-1,1)$ family rank $= \lceil (n-2)/2 \rceil$. A unifying combinatorial formula for λ is open.

4. **Connection to σ_1.** Empirically, at every λ tested, $\sigma_1(V_\lambda) = 0 \iff \operatorname{rank}\Pi = 0$. The forward direction (rank 0 $\Rightarrow$ trace 0) is trivial. The reverse (trace 0 $\Rightarrow$ rank 0) is not immediate from idempotence considerations: while $(T_i + 1)^2 = (1 + q)(T_i + 1)$, the product $\Pi^{S_n}/(1 + q)^L$ is *not* an idempotent (the elements $\pi_i = (T_i + 1)/(1 + q)$ do not satisfy braid relations), so the trace–rank identity for idempotents does not apply. A separate proof is needed.

7 Honest assessment

What this paper proves.

1. Theorem 13: $\Pi^{S_n}|_{V_\lambda} = 0$ for all $\lambda \vdash n$ with $\ell(\lambda) > \lceil n/2 \rceil$ and all $n \leq 7$.
2. The clean inductive case (Corollary 5): the same vanishing for all n in the deep regime $\ell(\lambda) \geq \lceil n/2 \rceil + 2$, and for all n odd (provided the result holds at $n - 1$).

What is conjectural. The boundary lemma (Lemma 6) for $n \geq 8$ even. The reverse direction (Conjecture 14) for $n \geq 8$.

The gap. The induction-from-below approach reduces the problem to the boundary lemma. A general proof of the boundary lemma — giving either an explicit formula for $j(\lambda)$ or a basis-free characterization of when $B_j(V_\lambda) \subseteq E_1^-$ holds — is the natural next target. The structural ingredient appears to be the interplay of T_1-eigenspaces with the iterated R'_k-action; this is essentially the same flavor of analysis as in the May-7 rank-1 writeup (`2026-05-08-rank-Pi-structural.tex`), but driving the image into E_1^- rather than into a specific 1-dimensional line.

Why the staircase, not a general reduced word. Different reduced expressions of w_0 give Bott–Samelson elements that differ as operators (the $\pi_i = (T_i + 1)/(1 + q)$ do not satisfy braid relations). The staircase has a particularly clean recursive structure $\Pi^{S_n} = \Pi^{S_{n-1}} R'_n$ which drives the inductive proof. We do not assert the rank-zero characterization for other reduced words.

Computational note. All matrix computations are in the seminormal basis using SymPy with rational $q \in \mathbb{Q}$. The rank survey at $n = 7$ uses numerical evaluation in the KL basis at integer q-values. Scripts:

`~/projects/scratch/2026-05-08-rank-0-Pi/test_rank0_claim.py`,
`~/projects/scratch/2026-05-08-rank-0-Pi/test_iterated_claim.py`,
`~/projects/scratch/2026-05-08-rank-0-Pi/rank_n7_survey.py`.